

8 (short)

(a) Explain the difference between File Register Instructions and Literal Instructions in PIC programs. [2]

(b) The PIC code below is executed.

```
movlw    15;  
movwf    0x20;  
movlw    0xF1;  
addwf    0x20,W;  
movlw    0xAA;  
movwf    0x20;  
movlw    255;  
xorwf    0x20;  
sleep;
```

Determine the contents of W, memory location 0x20 and the C, DC and Z flags after:

(i) the **addwf 0x20, W** instruction; [4]

(ii) the **xorwf 0x20** instruction. [4]